MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati
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Lectures 15-16: Systems of first-order equations, Matrix form of first-order systems

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Attention students: Please keep in mind that it is not always possible to discuss all problems during the tutorial. You can consider the problems not discussed as your practice problems.

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1. Represent each of the following equations in terms of an equivalent system of first-order equations:
   (a) $x'' - t^2x' - tx = 0$, (b) $x''' = x'' - t^2(x')^2$, (c) $x'' - 3x' - 11x = \sin t$.
   Further, write the derived systems in matrix form $\bar{x}' = A\bar{x} + \bar{b}$.

2. If a particle of mass moves in the xy-plane, its equations of motion are
$$m\frac{d^2x}{dt^2} = f(t, x, y), \quad m\frac{d^2y}{dt^2} = g(t, x, y),$$
   where f and g represent the x and y components, respectively, of the force acting on the particle. Express this system of two second-order equations by an equivalent system of four first-order equations.

3. The vector functions $\bar{x}_1 = [e^{-t}, 2e^{-t}, e^{-t}]^T$, $\bar{x}_2 = [e^t, 0, e^t]^T$, $\bar{x}_3 = [e^{3t}, -e^{3t}, 2e^{3t}]^T$ are solutions to the system $\bar{x}'(t) = A\bar{x}(t)$. Determine whether they form a fundamental solution set. If they do, find a fundamental matrix for the system and give a general solution.

4. Find the general solution of the following first-order equations:
$$\begin{cases} \frac{dx}{dt} = 4x - y, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

5. Find the general solution of the following first-order equations:
$$\begin{cases} \frac{dx}{dt} = x + 3y, \\ \frac{dy}{dt} = 3x + y. \end{cases}$$
   Then, find the particular solution given that $x(0) = 5$ and $y(0) = 1$.

6. Solve the given equation $\bar{x}'(t) = A\bar{x}(t)$ for
$$\text{(a) } A = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \text{ (b) } A = \begin{bmatrix} -1 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & 1 \end{bmatrix}.$$

7. Show that the solutions for the system

$$\begin{cases} \frac{dx}{dt} = x + 2y, \\ \frac{dy}{dt} = 3x + 2y. \end{cases}$$

are given by $\begin{cases} x = 2e^{4t}, \\ y = 3e^{4t}, \end{cases}$ and $\begin{cases} x = e^{-t}, \\ y = -e^{-t}. \end{cases}$
Further, show that the solutions of the above system can also be obtained by
(i) differentiating the first equation with respect to t and then eliminating y,
(ii) differentiating the second equation with respect to t and then eliminating x.

8. We know that, for a system of two first-order equations, the complex roots $a \pm ib$ of the auxiliary equations lead to the following solutions:

$$\left.\begin{aligned} x &= e^{at}(A_1 \cos bt - A_2 \sin bt), \\ y &= e^{at}(B_1 \cos bt - B_2 \sin bt), \end{aligned}\right\}$$

and

$$\left.\begin{aligned} x &= e^{at}(A_1 \sin bt + A_2 \cos bt), \\ y &= e^{at}(B_1 \sin bt + B_2 \cos bt). \end{aligned}\right\}$$

Find the Wronskian of these solutions. Check whether it is mandatory that $A_1B_2 - A_2B_1 \neq 0$.

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